In [ ]:
```
EXPERIMENT 4 – DATA MANIPULATION WITH PANDAS
PIN NO:- 241UDS0111

AIM
To perform data manipulation operations using Pandas, including indexing, selecting rows and columns, creating
calculated columns, filtering, sorting, updating values, and renaming columns.
```

In [ ]:
```
## Step 1 – Install and Import Pandas Library
If Pandas is not installed, run `%pip install pandas` in a separate code cell first.
```

In [1]:
```
import pandas as pd
```

In [ ]:
```
## Step 2 – Importing the Dataset
```

In [2]:
```
# Importing the Dataset
df_241UDS0111 = pd.read_csv("Experiment_4_241UDS0111_Plant_Nursery_Sales.csv")

# Reading the head of the Dataset
df_241UDS0111.head(5)
```

Out[2]: Columns 1–8 of 11

	Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
0	N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0
1	N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0
2	N2003	Terracotta Pot	Pots	Visakhapatnam	44	180	6.0
3	N2004	Jade Plant	Succulents	Pune	27	350	10.0
4	N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0

Columns 9–11 of 11 · first column repeated

	Customer_Rating	Stock_Available	Payment_Method
0	4.6	21	UPI
1	4.4	52	Cash
2	4.2	38	Card
3	4.7	17	UPI
4	4.1	46	Card

In [ ]:
```
## Step 3 – Set Index for the DataFrame
```

In [3]:
```
# Set Index for the DataFrame
df_241UDS0111 = df_241UDS0111.set_index("Plant_ID")
df_241UDS0111.head()
```

Out[3]: Columns 1–8 of 10

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4
N2003	Terracotta Pot	Pots	Visakhapatnam	44	180	6.0	4.2
N2004	Jade Plant	Succulents	Pune	27	350	10.0	4.7
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1

Columns 9–10 of 10 · first column repeated

Plant_ID	Stock_Available	Payment_Method
N2001	21	UPI
N2002	52	Cash
N2003	38	Card
N2004	17	UPI
N2005	46	Card

In [ ]:
```
## Step 4 – Info of the DataFrame
```

In [4]:
```
df_241UDS0111.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, N2001 to N2030
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Item_Name         30 non-null     object 
 1   Category          30 non-null     object 
 2   City              30 non-null     object 
 3   Units_Sold        30 non-null     int64  
 4   Unit_Price        30 non-null     int64  
 5   Discount_Percent  30 non-null     float64
 6   Customer_Rating   30 non-null     float64
 7   Stock_Available   30 non-null     int64  
 8   Payment_Method    30 non-null     object 
dtypes: float64(2), int64(3), object(4)
memory usage: 2.3+ KB
```

In [ ]:
```
## Step 5 – Columns & Rows Selection
```

In [5]:
```
# Selecting a Single Column
df_241UDS0111["Item_Name"]
```

Out[5]:
```
Plant_ID
N2001            Snake Plant
N2002         Basil Seedling
N2003         Terracotta Pot
N2004             Jade Plant
N2005         Cocopeat Block
N2006     Marigold Seed Pack
N2007             Peace Lily
N2008          Mint Seedling
N2009        Ceramic Planter
N2010         Pruning Shears
N2011    Sunflower Seed Pack
N2012         Monstera Plant
N2013              Aloe Vera
N2014          Rose Seedling
N2015          Garden Gloves
N2016      Lavender Seedling
N2017       Zinnia Seed Pack
N2018         Hanging Basket
N2019           Rubber Plant
N2020       Vermicompost Bag
N2021             Cactus Mix
N2022       Jasmine Seedling
N2023           Neem Sapling
N2024           Orchid Plant
N2025       Tomato Seed Pack
N2026           Watering Can
N2027             Areca Palm
N2028         Wooden Planter
N2029       Petunia Seedling
N2030          Garden Trowel
Name: Item_Name, dtype: object
```

In [ ]:
```
## Step 6 – Selecting Multiple Columns
```

In [6]:
```
# Selecting Multiple Columns
df_241UDS0111[["Item_Name", "Category", "City"]].head(3)
```

Out[6]:

	Item_Name	Category	City
Plant_ID			
N2001	Snake Plant	Indoor Plants	Hyderabad
N2002	Basil Seedling	Herbs	Vijayawada
N2003	Terracotta Pot	Pots	Visakhapatnam

In [ ]:
```
## Step 7 – Select Row Using iloc
```

In [7]:
```
# Select by row index using iloc (position-based)
df_241UDS0111.iloc[1]
```

Out[7]:
```
Item_Name           Basil Seedling
Category                     Herbs
City                     Vijayawada
Units_Sold                       68
Unit_Price                       90
Discount_Percent                8.0
Customer_Rating                 4.4
Stock_Available                  52
Payment_Method                 Cash
Name: N2002, dtype: object
```

In [ ]:
```
## Step 8 — Select Row Using loc
```

In [8]:
```
# Select by row index using loc (label-based)
df_241UDS0111.loc["N2030"]
```

Out[8]:
```
Item_Name               Garden Trowel
Category            Gardening Supplies
City                         Hyderabad
Units_Sold                          19
Unit_Price                         285
Discount_Percent                  15.0
Customer_Rating                    4.5
Stock_Available                     10
Payment_Method                     UPI
Name: N2030, dtype: object
```

In [ ]:
```
## Step 9 — Manipulating Data Based on Columns
```

In [9]:
```
# Display first two rows
df_241UDS0111.head(2)
```

Out[9]:

Columns 1–8 of 10

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–10 of 10 · first column repeated

Plant_ID	Stock_Available	Payment_Method
N2001	21	UPI
N2002	52	Cash

In [ ]:
```
## Step 10 — Creating Total Sales Column
```

In [10]:
```
# Creating Total Sales Column
df_241UDS0111["Total_Sales_241UDS0111"] = (
    df_241UDS0111["Units_Sold"] * df_241UDS0111["Unit_Price"]
)
df_241UDS0111.head(2)
```

Out[10]: Columns 1–8 of 11

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–11 of 11 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111
N2001	21	UPI	13440
N2002	52	Cash	6120

In [ ]:
```
## Step 11 — Creating Discount Amount Column
```

In [11]:
```
# Creating Discount Amount Column
df_241UDS0111["Discount_Amount_241UDS0111"] = (
    df_241UDS0111["Total_Sales_241UDS0111"] *
    df_241UDS0111["Discount_Percent"] / 100
)
df_241UDS0111.head(2)
```

Out[11]: Columns 1–8 of 12

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–12 of 12 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111
N2001	21	UPI	13440	672.0
N2002	52	Cash	6120	489.6

In [ ]:
```
## Step 12 — Creating Net Sales Column
```

In [12]:
```
# Creating Net Sales Column
df_241UDS0111["Net_Sales_241UDS0111"] = (
    df_241UDS0111["Total_Sales_241UDS0111"] -
    df_241UDS0111["Discount_Amount_241UDS0111"]
)
df_241UDS0111.head(2)
```

Out[12]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2002	52	Cash	6120	489.6	5630.4

In [ ]:
```
## Step 13 — Round Calculated Values
```

In [13]:
```
# Round calculated columns to 2 decimal places
df_241UDS0111["Discount_Amount_241UDS0111"] = df_241UDS0111["Discount_Amount_241UDS0111"].round(2)
df_241UDS0111["Net_Sales_241UDS0111"] = df_241UDS0111["Net_Sales_241UDS0111"].round(2)
df_241UDS0111.head(2)
```

Out[13]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2002	52	Cash	6120	489.6	5630.4

In [ ]:
```
## Step 14 — Conditional Filtering
```

In [14]:
```
# Items having Units Sold greater than 50
df_241UDS0111[df_241UDS0111["Units_Sold"] > 50].head(2)
```

Out[14]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2002	52	Cash	6120	489.6	5630.4
N2005	46	Card	7420	519.4	6900.6

In [ ]:
```
## Step 15 — Products with High Sales AND Low Stock
```

In [15]:
```
# Items with Total Sales > 10000 AND Stock Available < 25
df_241UDS0111[
    (df_241UDS0111["Total_Sales_241UDS0111"] > 10000) &
    (df_241UDS0111["Stock_Available"] < 25)
].head(3)
```

Out[15]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2007	Peace Lily	Indoor Plants	Bengaluru	24	580	5.0	4.8
N2009	Ceramic Planter	Pots	Hyderabad	29	760	11.0	4.7

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2007	14	UPI	13920	696.0	13224.0
N2009	18	UPI	22040	2424.4	19615.6

In [ ]:
```
## Step 16 — Products from Hyderabad OR Mumbai
```

In [16]:
```
# Items from Hyderabad or Mumbai
df_241UDS0111[
    (df_241UDS0111["City"] == "Hyderabad") |
    (df_241UDS0111["City"] == "Mumbai")
].head()
```

Out[16]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1
N2009	Ceramic Planter	Pots	Hyderabad	29	760	11.0	4.7
N2013	Aloe Vera	Succulents	Mumbai	48	240	7.0	4.5
N2016	Lavender Seedling	Herbs	Hyderabad	39	275	9.0	4.6

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.00	12768.00
N2005	46	Card	7420	519.40	6900.60
N2009	18	UPI	22040	2424.40	19615.60
N2013	35	Card	11520	806.40	10713.60
N2016	26	Cash	10725	965.25	9759.75

In [ ]:
```
## Step 17 — Products with Rating ≥ 4.5 AND Stock < 30
```

In [17]:
```
# Items with high customer rating and low stock
df_241UDS0111[
    (df_241UDS0111["Customer_Rating"] >= 4.5) &
    (df_241UDS0111["Stock_Available"] < 30)
]
```

Out[17]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2004	Jade Plant	Succulents	Pune	27	350	10.0	4.7
N2007	Peace Lily	Indoor Plants	Bengaluru	24	580	5.0	4.8
N2009	Ceramic Planter	Pots	Hyderabad	29	760	11.0	4.7
N2012	Monstera Plant	Indoor Plants	Pune	34	890	10.0	4.6
N2016	Lavender Seedling	Herbs	Hyderabad	39	275	9.0	4.6
N2018	Hanging Basket	Pots	Delhi	31	410	12.0	4.8
N2019	Rubber Plant	Indoor Plants	Kochi	22	670	8.0	4.7
N2022	Jasmine Seedling	Flowering Plants	Mumbai	35	295	10.0	4.5
N2024	Orchid Plant	Indoor Plants	Delhi	41	720	9.0	4.7
N2026	Watering Can	Gardening Supplies	Mumbai	33	250	7.0	4.6
N2027	Areca Palm	Outdoor Plants	Bengaluru	27	640	11.0	4.8
N2030	Garden Trowel	Gardening Supplies	Hyderabad	19	285	15.0	4.5

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.00	12768.00
N2004	17	UPI	9450	945.00	8505.00
N2007	14	UPI	13920	696.00	13224.00
N2009	18	UPI	22040	2424.40	19615.60
N2012	22	UPI	30260	3026.00	27234.00
N2016	26	Cash	10725	965.25	9759.75
N2018	19	Card	12710	1525.20	11184.80
N2019	13	UPI	14740	1179.20	13560.80
N2022	24	UPI	10325	1032.50	9292.50
N2024	25	Card	29520	2656.80	26863.20
N2026	15	Card	8250	577.50	7672.50
N2027	18	UPI	17280	1900.80	15379.20
N2030	10	UPI	5415	812.25	4602.75

In [ ]:
```
## Step 18 — Filtering Multiple Cities Using .isin()
```

In [18]:
```
# Filtering multiple cities using .isin()
df_241UDS0111[
    df_241UDS0111["City"].isin(["Hyderabad", "Bengaluru", "Pune"])
].head()
```

Out[18]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2004	Jade Plant	Succulents	Pune	27	350	10.0	4.7
N2007	Peace Lily	Indoor Plants	Bengaluru	24	580	5.0	4.8
N2009	Ceramic Planter	Pots	Hyderabad	29	760	11.0	4.7
N2012	Monstera Plant	Indoor Plants	Pune	34	890	10.0	4.6

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2004	17	UPI	9450	945.0	8505.0
N2007	14	UPI	13920	696.0	13224.0
N2009	18	UPI	22040	2424.4	19615.6
N2012	22	UPI	30260	3026.0	27234.0

In [ ]:
```
## Step 19 — Filtering Multiple Categories Using .isin()
```

In [19]:
```
# Filtering multiple categories using .isin()
df_241UDS0111[
    df_241UDS0111["Category"].isin(["Indoor Plants", "Gardening Supplies"])
].head()
```

Out[19]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1
N2007	Peace Lily	Indoor Plants	Bengaluru	24	580	5.0	4.8
N2010	Pruning Shears	Gardening Supplies	Kochi	37	320	6.0	4.2
N2012	Monstera Plant	Indoor Plants	Pune	34	890	10.0	4.6

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2005	46	Card	7420	519.4	6900.6
N2007	14	UPI	13920	696.0	13224.0
N2010	28	Card	11840	710.4	11129.6
N2012	22	UPI	30260	3026.0	27234.0

In [ ]:
```
## Step 20 — Filtering Multiple Payment Methods Using .isin()
```

In [20]:
```
# Filtering multiple payment methods using .isin()
df_241UDS0111[
    df_241UDS0111["Payment_Method"].isin(["UPI", "Card"])
].head()
```

Out[20]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2001	Snake Plant	Indoor Plants	Hyderabad	32	420	5.0	4.6
N2003	Terracotta Pot	Pots	Visakhapatnam	44	180	6.0	4.2
N2004	Jade Plant	Succulents	Pune	27	350	10.0	4.7
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1
N2007	Peace Lily	Indoor Plants	Bengaluru	24	580	5.0	4.8

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2001	21	UPI	13440	672.0	12768.0
N2003	38	Card	7920	475.2	7444.8
N2004	17	UPI	9450	945.0	8505.0
N2005	46	Card	7420	519.4	6900.6
N2007	14	UPI	13920	696.0	13224.0

In [ ]:
```
## Step 21 — Sorting Table
```

In [21]:
```
# Sorting table based on Units Sold (descending)
df_241UDS0111.sort_values("Units_Sold", ascending=False).head()
```

Out[21]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2006	Marigold Seed Pack	Seeds	Delhi	96	55	12.0	4.5
N2017	Zinnia Seed Pack	Seeds	Vijayawada	91	48	5.0	4.0
N2011	Sunflower Seed Pack	Seeds	Delhi	82	65	8.0	4.4
N2025	Tomato Seed Pack	Seeds	Hyderabad	74	42	5.0	4.1
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2006	70	Cash	5280	633.6	4646.4
N2017	63	UPI	4368	218.4	4149.6
N2011	57	Cash	5330	426.4	4903.6
N2025	49	UPI	3108	155.4	2952.6
N2002	52	Cash	6120	489.6	5630.4

In [ ]:
```
## Step 22 — Sort by Unit Price — Ascending
```

In [22]:
```
# Sorting based on Unit Price (ascending)
df_241UDS0111.sort_values("Unit_Price", ascending=True).head()
```

Out[22]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2025	Tomato Seed Pack	Seeds	Hyderabad	74	42	5.0	4.1
N2017	Zinnia Seed Pack	Seeds	Vijayawada	91	48	5.0	4.0
N2006	Marigold Seed Pack	Seeds	Delhi	96	55	12.0	4.5
N2011	Sunflower Seed Pack	Seeds	Delhi	82	65	8.0	4.4
N2008	Mint Seedling	Herbs	Chennai	61	85	9.0	4.3

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2025	49	UPI	3108	155.40	2952.60
N2017	63	UPI	4368	218.40	4149.60
N2006	70	Cash	5280	633.60	4646.40
N2011	57	Cash	5330	426.40	4903.60
N2008	41	Cash	5185	466.65	4718.35

In [ ]:
```
## Step 23 — Sorting Using Multiple Columns
```

In [23]:
```
# Sorting based on Category and Units Sold
df_241UDS0111.sort_values(
    ["Category", "Units_Sold"],
    ascending=[True, False]
).head()
```

Out[23]: Columns 1–8 of 13

Plant_ID	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
N2029	Petunia Seedling	Flowering Plants	Kochi	63	115	6.0	4.2
N2014	Rose Seedling	Flowering Plants	Bengaluru	57	160	6.0	4.3
N2022	Jasmine Seedling	Flowering Plants	Mumbai	35	295	10.0	4.5
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1
N2020	Vermicompost Bag	Gardening Supplies	Visakhapatnam	46	190	7.0	4.4

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
N2029	45	Card	7245	434.7	6810.3
N2014	43	UPI	9120	547.2	8572.8
N2022	24	UPI	10325	1032.5	9292.5
N2005	46	Card	7420	519.4	6900.6
N2020	32	Cash	8740	611.8	8128.2

In [ ]:
```
## Step 24 — Updating Data in the Table
```

In [24]:
```
# Updating Hyderabad as Hyd in City Column
df_241UDS0111["City"] = df_241UDS0111["City"].replace(["Hyderabad"], ["Hyd"])
df_241UDS0111[df_241UDS0111["City"] == "Hyd"].head(2)
```

Out[24]: Columns 1–8 of 13

	Item_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Plant_ID							
N2001	Snake Plant	Indoor Plants	Hyd	32	420	5.0	4.6
N2009	Ceramic Planter	Pots	Hyd	29	760	11.0	4.7

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Net_Sales_241UDS0111
Plant_ID					
N2001	21	UPI	13440	672.0	12768.0
N2009	18	UPI	22040	2424.4	19615.6

In [ ]:
```
## Step 25 — Updating a Specific Product Value
```

In [25]:
```
# Updating Plant N2001 Unit Price
df_241UDS0111.loc["N2001", "Unit_Price"] = 460
print(df_241UDS0111.loc["N2001", "Unit_Price"])
```

```
460
```

In [ ]:
```
## Step 26 — Changing Column Names
```

In [26]:
```
# Changing selected column names
df_241UDS0111 = df_241UDS0111.rename(columns={
    "Item_Name": "Plant_Item",
    "City": "Location",
    "Customer_Rating": "Rating",
    "Net_Sales_241UDS0111": "Revenue"
})
df_241UDS0111.head(1)
```

Out[26]: Columns 1–8 of 13

	Plant_Item	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
Plant_ID							
N2001	Snake Plant	Indoor Plants	Hyd	32	460	5.0	4.6

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Revenue
Plant_ID					
N2001	21	UPI	13440	672.0	12768.0

In [ ]:
```
## Step 27 — Final DataFrame
```

In [27]:
```
# Display the final DataFrame
df_241UDS0111.head()
```

Out[27]: Columns 1–8 of 13

Plant_ID	Plant_Item	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
N2001	Snake Plant	Indoor Plants	Hyd	32	460	5.0	4.6
N2002	Basil Seedling	Herbs	Vijayawada	68	90	8.0	4.4
N2003	Terracotta Pot	Pots	Visakhapatnam	44	180	6.0	4.2
N2004	Jade Plant	Succulents	Pune	27	350	10.0	4.7
N2005	Cocopeat Block	Gardening Supplies	Mumbai	53	140	7.0	4.1

Columns 9–13 of 13 · first column repeated

Plant_ID	Stock_Available	Payment_Method	Total_Sales_241UDS0111	Discount_Amount_241UDS0111	Revenue
N2001	21	UPI	13440	672.0	12768.0
N2002	52	Cash	6120	489.6	5630.4
N2003	38	Card	7920	475.2	7444.8
N2004	17	UPI	9450	945.0	8505.0
N2005	46	Card	7420	519.4	6900.6

In [ ]:
```
## Step 28 — Final Dataset Information
```

In [28]:
```
# Final information of the DataFrame
df_241UDS0111.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 30 entries, N2001 to N2030
Data columns (total 12 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Plant_Item                  30 non-null     object
 1   Category                    30 non-null     object
 2   Location                    30 non-null     object
 3   Units_Sold                  30 non-null     int64
 4   Unit_Price                  30 non-null     int64
 5   Discount_Percent            30 non-null     float64
 6   Rating                      30 non-null     float64
 7   Stock_Available             30 non-null     int64
 8   Payment_Method              30 non-null     object
 9   Total_Sales_241UDS0111      30 non-null     int64
 10  Discount_Amount_241UDS0111  30 non-null     float64
 11  Revenue                     30 non-null     float64
dtypes: float64(4), int64(4), object(4)
memory usage: 4.1+ KB
```

In [ ]:
```
RESULT:-
Using Pandas, data manipulation techniques including row and column selection, indexing, calculated columns,
conditional filtering, sorting, updating values, and renaming columns were performed on the plant-nursery sales
dataset
```